

Agnos Candidate Assignment

Each candidate will need to complete the assignment within 3 days of receiving the task.

Back-end developer:

Tasks:

1. Develop APIs for a Hospital Middleware system to search and display patient information from Hospital Information Systems (HIS) using the following details:

Hospital A API:

- Route: GET <https://hospital-a.api.co.th/patient/search/{id}>
 - Request Parameters:
 - id (string): Can be either national_id or passport_id
 - Response Body (JSON):
 - first_name_th
 - middle_name_th
 - last_name_th
 - first_name_en
 - middle_name_en
 - last_name_en
 - date_of_birth
 - patient_hn
 - national_id
 - passport_id
 - phone_number
 - email
 - gender (M, F)
2. Design a database schema for a "Patient" model that is compatible with hospital data structures.
 3. Design a database schema for a "Staff" model where each staff can only search for patient in the same hospital as the staff
 4. Implement the following APIs:
 - API to create a new hospital staff member with login credentials (/staff/create):
 - Input: username, password, hospital
 - API for staff login (/staff/login):
 - Input: username, password, hospital
 - API to search for a patient (/patient/search):
 - Requires login
 - Input: (all fields are optional)

- national_id
 - passport_id
 - first_name
 - middle_name
 - last_name
 - date_of_birth
 - phone_number
 - email
 - Output: Patients matching the criteria and belonging to the same hospital as the staff member
5. Create unit tests for each API, covering both positive and negative test scenarios.

Tech Stack:

1. Go
2. Gin Framework
3. Docker
4. Nginx
5. Postgres

*The resources are proprietary and are intellectual property of Agnos health co. ltd.
DO NOT SHARE WITH OTHERS*

Deliverables:

1. Development planning documentations share as google doc
 - a. Project Structure
 - b. API Spec
 - c. ER Diagram
2. Setup server with docker compose contains nginx, golang service, postgresql
3. Share your github

Evaluation Criteria:

1. **Requirement satisfaction**
2. **Code quality** (readability, maintainability, structure),
3. **Unit test coverage**
4. **Documentation clarity**